

1. O/N/23/22/Q3

A plane, irregular lamina has a mass of 50 g. It hangs from a nail that passes through a small hole H near to the edge of the lamina.

The nail acts as a pivot and the lamina can swing about it. The lamina is held in the position shown in Fig. 3.1, a small distance above a horizontal bench.

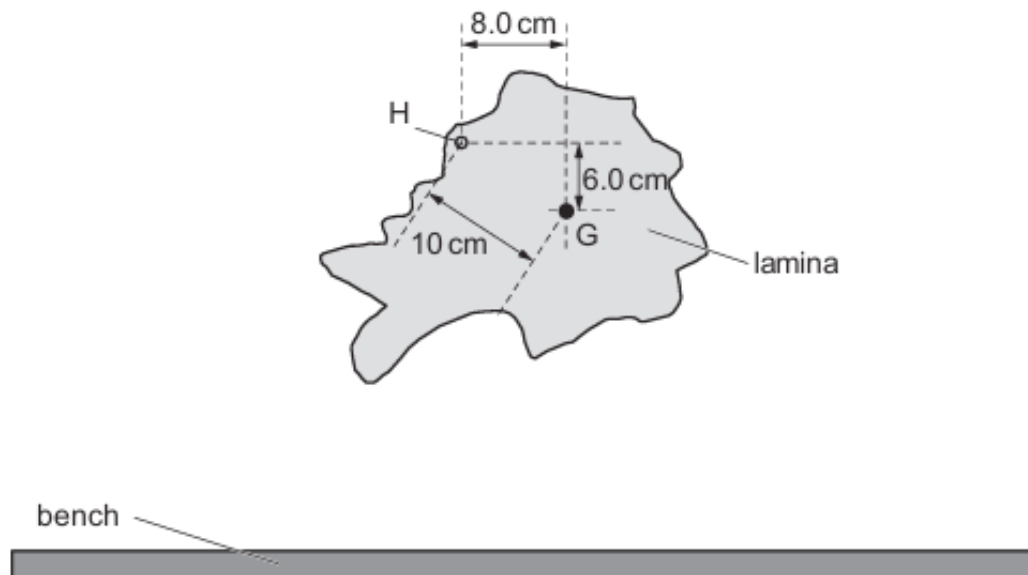


Fig. 3.1 (not to scale)

The centre of gravity of the lamina is at G.

(a) The weight of the lamina is a force that acts downwards.

(i) Explain why the lamina experiences this downward force.

.....
..... [1]

(ii) Calculate the weight of the lamina on Earth.

weight = N [2]

(b) The weight produces a moment about the pivot through H.

(i) Using Fig. 3.1, calculate the moment of the weight about the pivot.

moment = Nm [2]

(ii) The lamina is now released from the position shown in Fig. 3.1.

Describe what happens to the lamina from the time it is released until it stops moving.

.....

.....

.....

..... [3]

[Total: 8]

2. M/J/22/21/Q1

Fig. 1.1 shows a model of the human arm. The rubber band represents the muscle that moves part of the arm XY up.

A mass is suspended from XY, as shown in Fig. 1.2. The weight of section XY is negligible and the model is at rest.

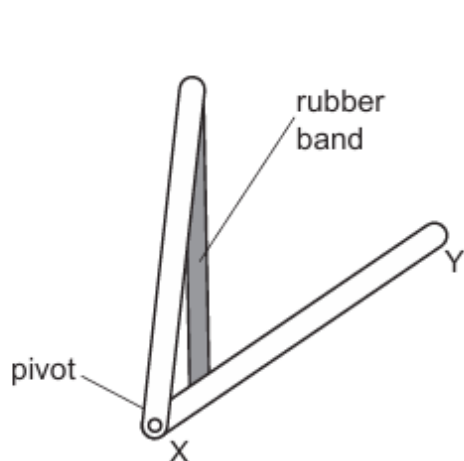


Fig. 1.1

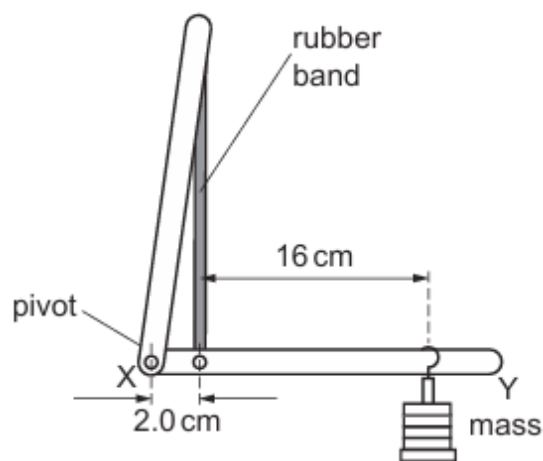


Fig. 1.2 (not to scale)

- (a) (i) State **two** ways in which the dimensions of the rubber band change as the mass is added to section XY.

.....
 [2]

- (ii) State the form of energy stored in the stretched rubber band.

..... [1]

- (b) (i) State the principle of moments.

.....

 [2]

- (ii) Explain why the force that the rubber band exerts on section XY is larger than the weight of the mass.

.....
.....
..... [1]

- (iii) The mass suspended from section XY in Fig. 1.2 has a weight of 4.0 N.

Calculate the force that the rubber band exerts on section XY.

force = [2]

- (iv) Explain how your answer to (b)(iii) is different if the weight of section XY is **not** negligible.

.....
.....
..... [1]

[Total: 9]

3. O/N/21/21/Q1

Fig. 1.1 shows a wooden bench of weight 2000 N.

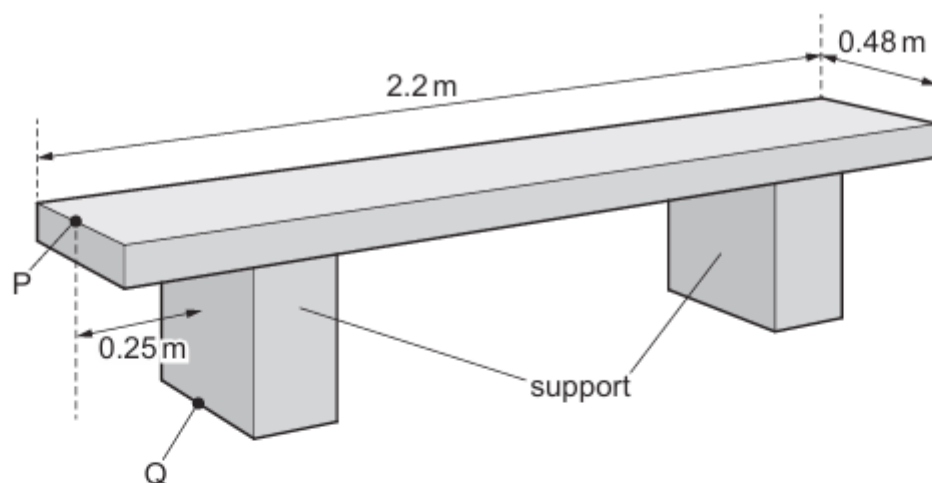


Fig. 1.1

- (a) Each of the two supports has an area of 0.040 m^2 in contact with the ground.

Calculate the pressure on the ground due to the bench.

pressure = [2]

- (b) The centre of mass of the bench is 1.1 m from the left-hand end of the bench and 0.24 m from the front.

- (i) Suggest **one** reason why the centre of mass is in this position.

.....

 [2]

- (ii) There is a force exerted vertically downwards from the point P shown in Fig. 1.1.

Calculate the maximum force that can be exerted vertically downwards at P without the bench rotating about the point Q shown in Fig. 1.1.

maximum force = [3]

[Total: 7]

4. O/N/21/22/Q8

The turning effect of a force is measured by its moment.

- (a) Fig. 8.1 shows a force F acting on an object at point P. The object is free to rotate about an axis at X that is perpendicular to the page.

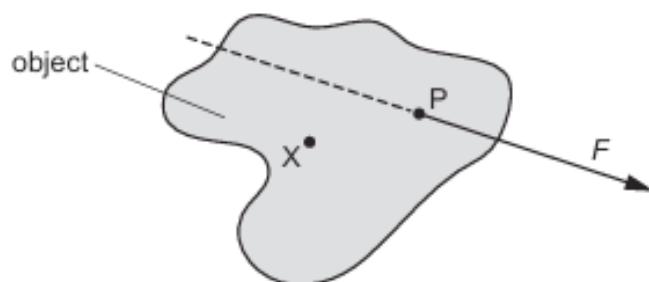


Fig. 8.1

Write down an expression for the moment of F about the axis at X. Draw on Fig. 8.1 to show what is meant by any other term used in your expression.

.....
 [2]

- (b) (i) State the principle of moments.

.....

 [2]

- (ii) Describe an experiment to verify the principle of moments. Include a diagram to help the description.

.....

 [3]

- (c) A worker carries a ladder on his shoulder. His shoulder acts as a pivot. Fig. 8.2 shows that the ladder is horizontal.

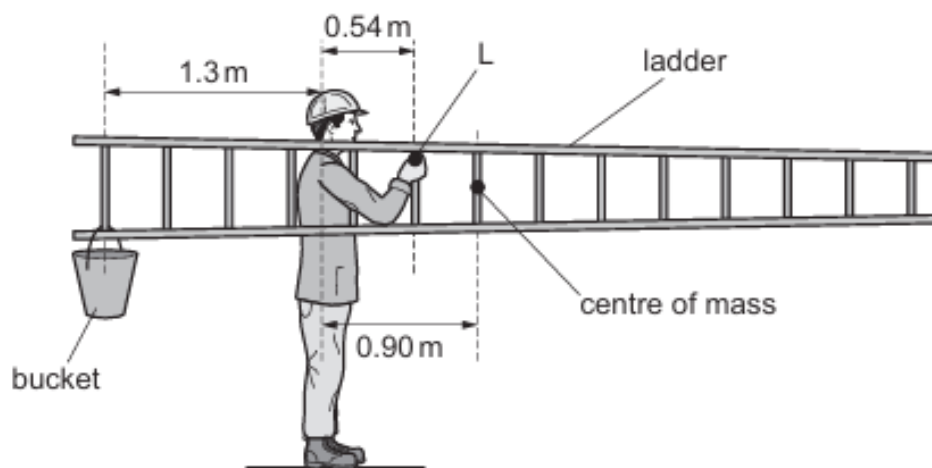


Fig. 8.2

The ladder is wider at one end than at the other end. The mass of the ladder is 8.0 kg.

- (i) The gravitational field strength is 10 N/kg.

Calculate the weight of the ladder.

weight = [1]

- (ii) State what is meant by *centre of mass*.

.....
 [1]

- (iii) The centre of mass of the ladder is not halfway along its length.

State what this shows about the ladder.

.....
 [1]

- (iv) The centre of mass of the ladder is a horizontal distance of 0.90 m from the worker's shoulder.

Calculate the moment about the worker's shoulder of the weight of the ladder.

moment = [2]

- (v) A bucket of weight 87 N is suspended from the ladder at a horizontal distance of 1.3 m from the worker's shoulder.

The worker keeps the ladder horizontal by exerting a vertical force at point L. L is a horizontal distance of 0.54 m from his shoulder.

Determine the size and direction of the force exerted at L.

size of force =

direction = [3]

[Total: 15]

5. M/J/21/22/Q3

Fig. 3.1 shows a small brick hanging from a newton meter.

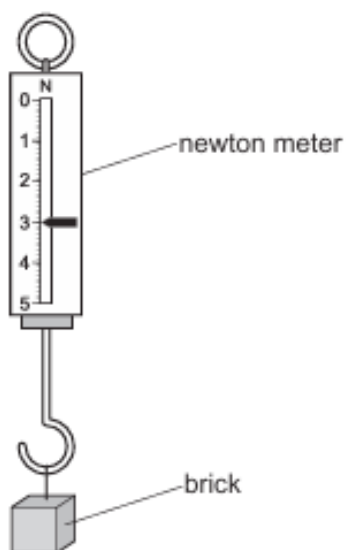


Fig. 3.1

The reading on the newton meter is 3.0 N.

- (a) Describe how the reading on the newton meter is used to find the mass of the brick.

.....
..... [1]

- (b) The same brick and newton meter are used in the apparatus shown in Fig. 3.2. The meter rule is pivoted at its centre and is balanced. The reading on the newton meter is not shown.

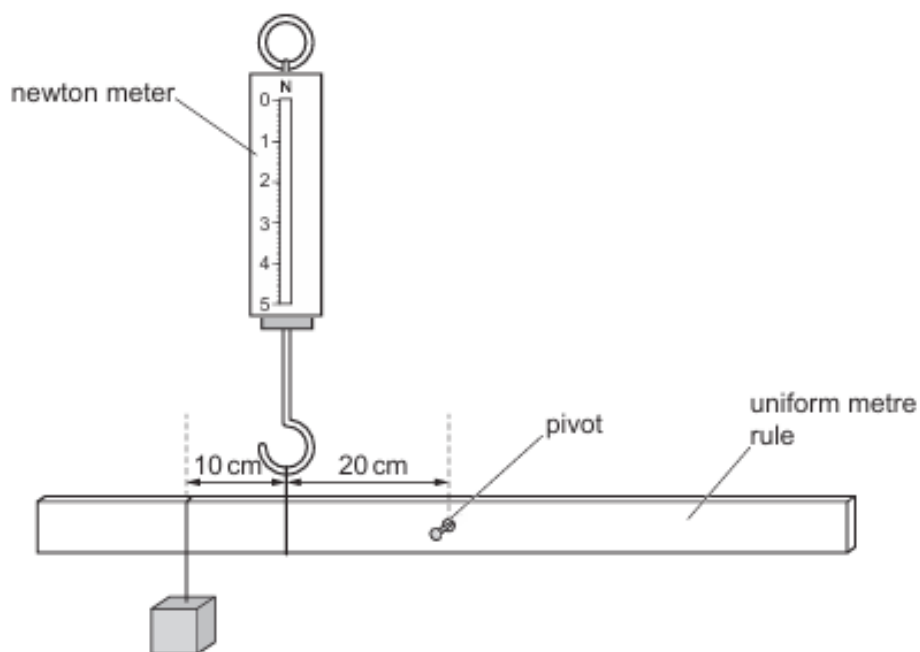


Fig. 3.2

- (i) State the *principle of moments* for a body in equilibrium.

.....

 [1]

- (ii) Determine the reading on the newton meter shown in Fig. 3.2.

reading = [2]

- (c) A beaker of water is placed so that the brick is partly submerged in the water, as shown in Fig. 3.3. The apparatus is adjusted to keep the rule horizontal.

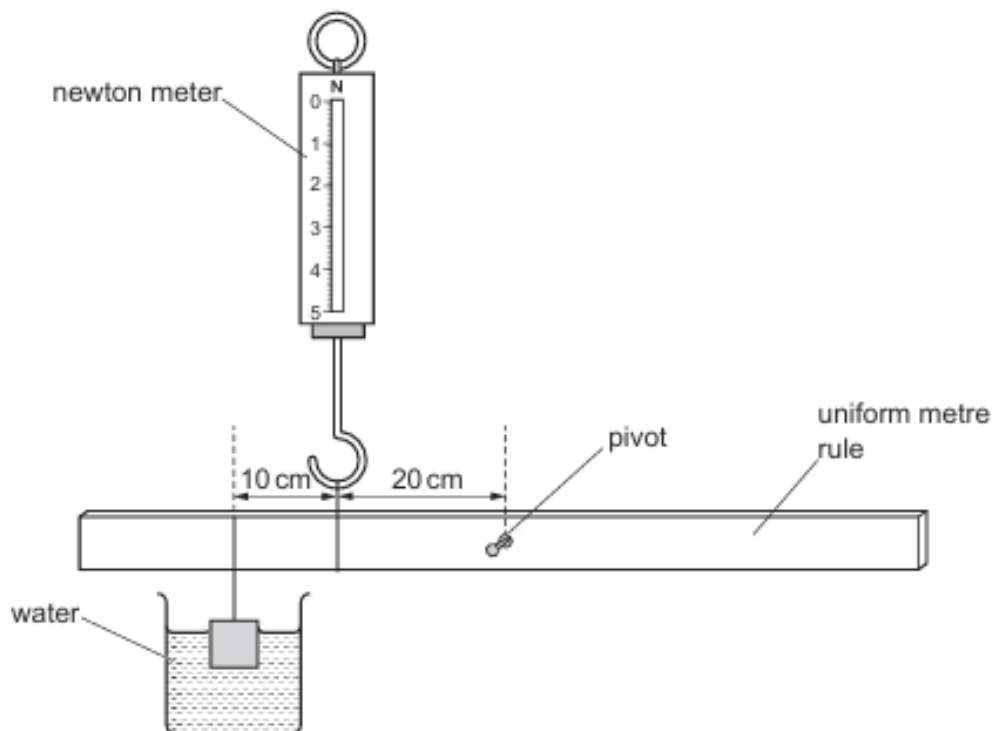


Fig. 3.3

Suggest why the reading on the newton meter is less than your answer in (b)(ii).

.....

 [2]

[Total: 6]

6. O/N/20/21/Q3

Fig. 3.1 shows a door and an automatic door-closer viewed from above.

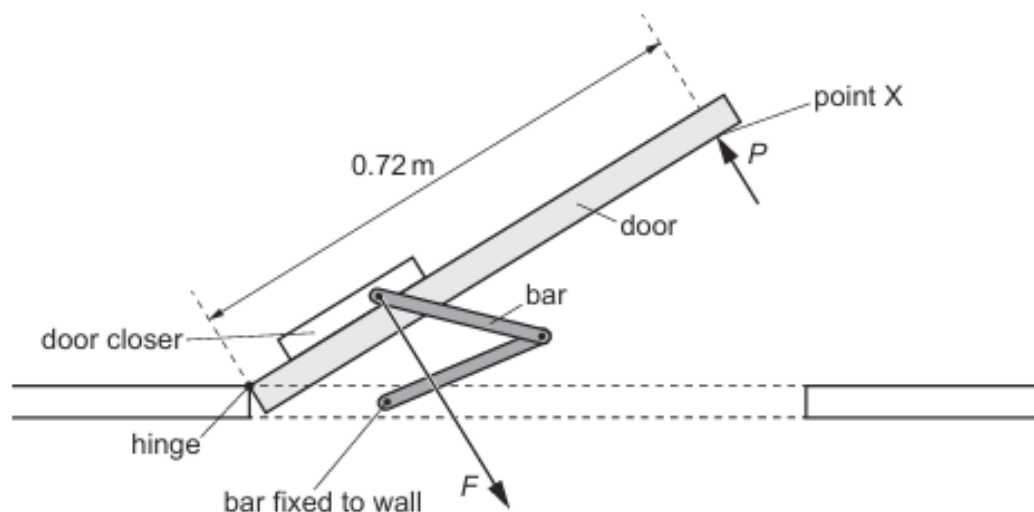


Fig. 3.1

When the door opens and closes, the hinge acts as a pivot.

A girl opens the door by exerting a force P at point X.

Force P is perpendicular to the surface of the door.

- (a) Fig. 3.1 shows that point X is a distance of 0.72 m along the front of the door from the hinge. The force P is 25 N.

- (i) Calculate the moment of force P about the hinge.

moment of force = [2]

- (ii) The door rotates about the hinge by 90° . The circumference of a circle of radius 0.72 m is 4.5 m.

Calculate the work done on the door by force P .

work done = [2]

(b) As the door opens, there is a force F on the door in the direction shown in Fig. 3.1.

Although force F is larger than force P , the door rotates about the hinge.

Explain why.

.....

.....

.....

..... [2]

[Total: 6]

7. O/N/20/22/Q1

A glass beaker has a mass of 50 g. A liquid of density 1.8 g/cm^3 is poured into the beaker until it reaches the 200 cm^3 mark.

- (a) Calculate the total mass of the beaker and its contents.

mass = [3]

- (b) The centre of mass of a metre rule is at the 50 cm mark.

- (i) State what is meant by *centre of mass*.

..... [1]

- (ii) The metre rule is placed on a pivot. The tip of the pivot is under the 80 cm mark on the rule.

The beaker with its contents is then placed at different positions along the rule until the rule is balanced.

Fig. 1.1 shows the arrangement with the rule balanced.

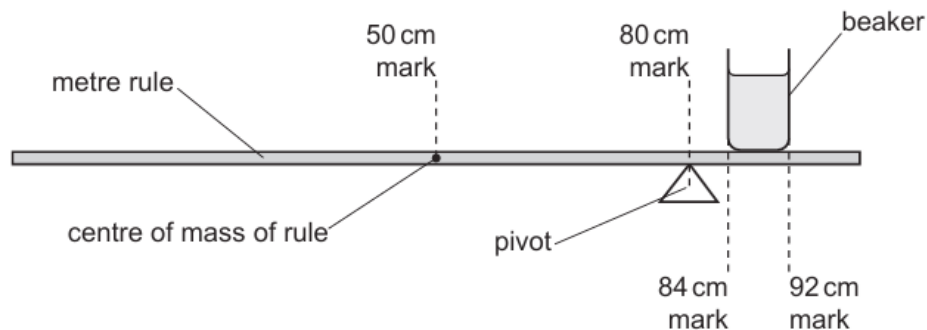


Fig. 1.1

One side of the beaker is at the 84 cm mark and the other side is at the 92 cm mark.

Calculate the mass of the rule.

mass = [3]

[Total: 7]

8. M/J/20/21/Q2

A student performs an experiment to mark the centre of mass C on a thin piece of card. There are two holes in the card.

Fig. 2.1 shows the card and two lines that the student draws on the card.

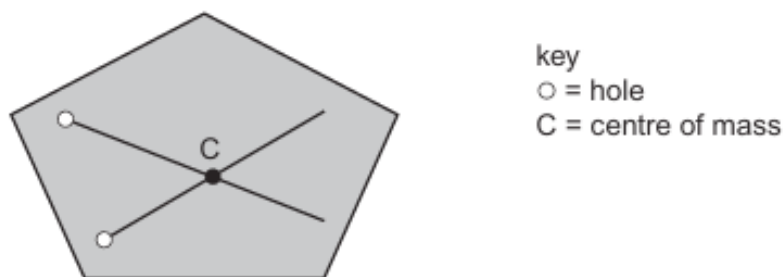


Fig. 2.1

- (a)** Describe a method used to draw these two lines in their correct positions on the card.

Make clear what extra apparatus is needed. You may draw a diagram, if you wish.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (b) The student holds the card loosely between her fingers. The card is vertical, resting with its lower edge on a bench as shown in Fig. 2.2.

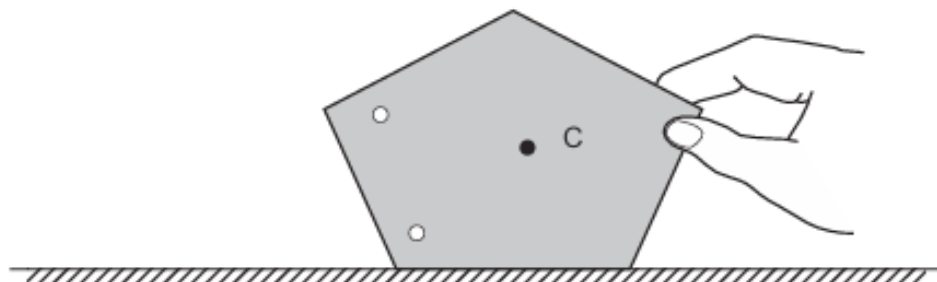


Fig. 2.2

The card is tilted slightly, as shown in Fig. 2.3, and then released.

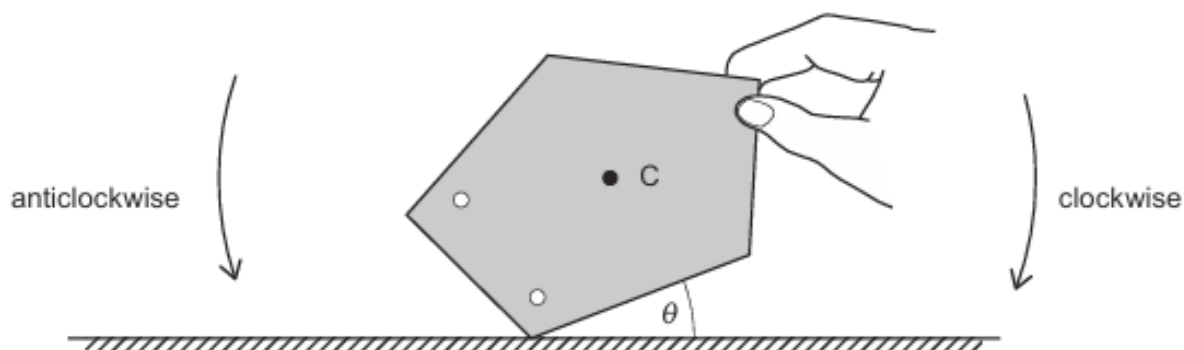


Fig. 2.3

When angle θ is small, the card falls clockwise, back to the position shown in Fig. 2.2.

- (i) Explain why the card falls anticlockwise when θ is large.

.....

.....

.....

..... [2]

- (ii) State one change to the card that makes it more stable.

.....

..... [1]

[Total: 6]

9. O/N/18/21/Q8

A bucket of water is pulled up out of a well using a rope. Fig. 8.1 shows the rope winding on to a cylinder as the handle is turned.

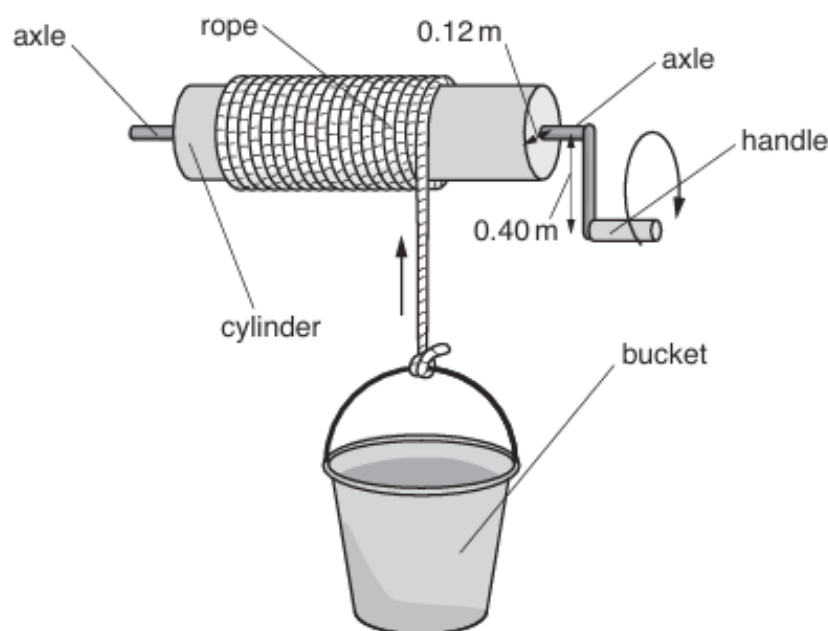


Fig. 8.1

The empty bucket has a mass of 1.0 kg.

- (a)** Complete the sentences that describe mass by filling in the gaps.

The mass of a body is a measure of the amount of
in the body. It resists a change in the state of of
the body. [2]

- (b)** When the bucket is full, it contains $2.4 \times 10^{-2} \text{ m}^3$ of water.

The gravitational field strength g is equal to 10 N/kg .

- (i)** Explain what is meant by a *gravitational field*.

.....
..... [1]

- (ii)** The density of water is 1000 kg/m^3 .

Determine the total weight of the bucket and the water.

weight = [3]

(c) The radius of the cylinder is 0.12 m and the handle is 0.40 m from the axle of the cylinder. The weight of the bucket and the water produce a moment that acts on the cylinder.

(i) Calculate this moment.

moment = [2]

(ii) Calculate the minimum force on the handle that balances this moment.

force = [1]

10. M/J/17/22/Q3

Fig. 3.1 shows the brake pedal of a car which is connected to a brake cylinder.

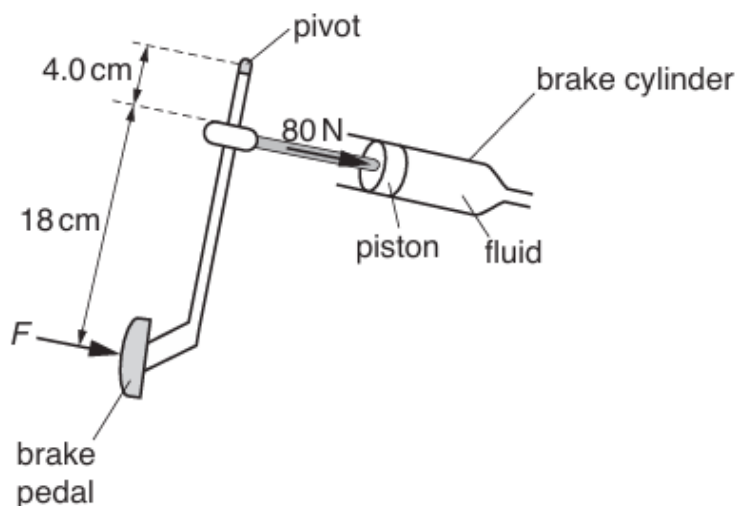


Fig. 3.1 (not to scale)

The brake is pressed with a force F . This force produces a moment about the pivot.

Pressing the brake causes a force of 80 N to act on the piston.

(a) Define the term *moment*.

.....
.....[2]

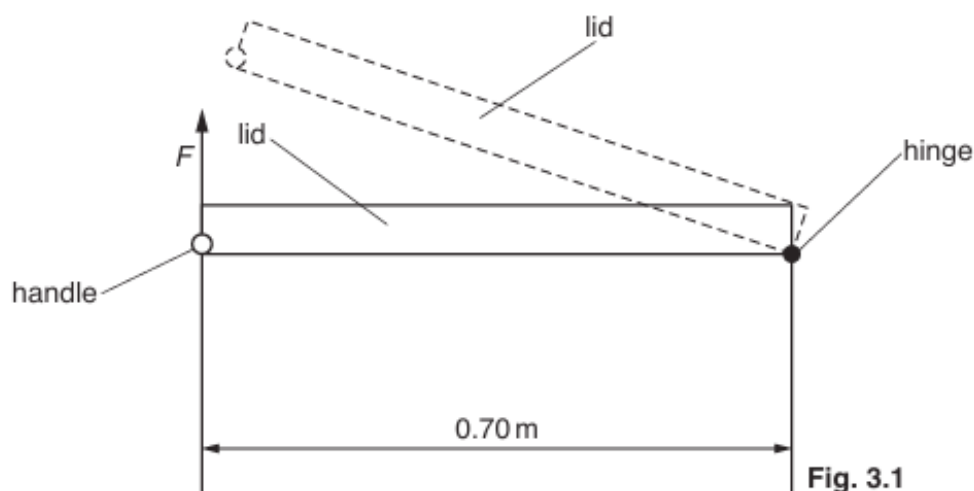
(b) Calculate the force F applied to the brake pedal.

$F =$ [2]

11. O/N/16/22/Q3

When the lid of a freezer is opened, it pivots about the hinge at the back of the freezer. The handle is at the front.

Fig. 3.1 is a side view of the freezer.



The handle is 0.70 m from the hinge. The lid has a mass of 2.0 kg.

- (a) The gravitational field strength g is 10 N/kg.

- (i) Calculate the weight of the lid.

weight =[1]

- (ii) The lid is uniform and its centre of mass is at its centre. The weight of the lid produces a moment about the hinge.

1. Calculate the moment about the hinge when the lid is closed.

moment =[2]

2. The moment required to open the lid is greater than the value calculated in (a)(ii)1.

Suggest one reason for this.

.....
[1]

- (b) The lid is closed. To open the lid, a force F is applied to the handle as shown in Fig. 3.1.

The direction of F is vertically upwards and F is the smallest possible force that opens the lid.

A force on the handle in any other direction must be larger than F in order to open the lid. Explain why.

.....
[1]

12. O/N/16/21/Q2

Fig. 2.1 shows a bed that folds away against a wall during the day.

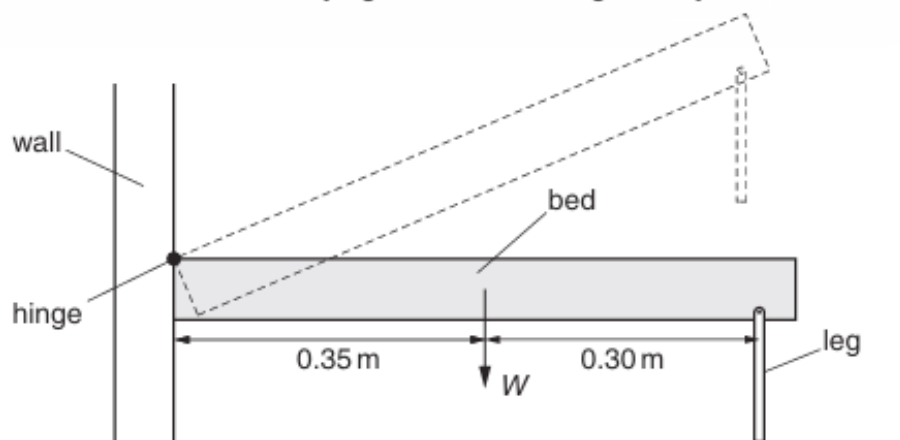


Fig. 2.1 (not to scale)

When it is horizontal, the bed is supported on one side by a hinge and on the other side by two legs. The weight W of the bed acts through its centre of mass, which is at a horizontal distance of 0.35 m from the hinge.

- (a) The mass of the bed is 26 kg. The gravitational field strength g is 10 N/kg.

Calculate the weight W of the bed.

$W = \dots\dots\dots$ [1]

- (b) (i) State the principle of moments.

.....

 [2]

- (ii) Determine the size of the total upward force exerted on the bed by the two legs when the bed is horizontal.

total force = [3]

13. O/N/15/21/Q4

Fig. 4.1 shows a screwdriver of mass 64g resting in equilibrium on a pivot.



Fig. 4.1

- (a) On Fig. 4.1, mark and label with a C, the centre of mass of the screwdriver. [1]
- (b) The gravitational field strength is 10 N/kg.
- (i) Calculate the weight of the screwdriver.

weight = [1]

- (ii) On Fig. 4.1, draw an arrow labelled *W* to represent the weight of the screwdriver. [1]

- (c) State two conditions that apply when an object is in equilibrium.

1.
2.

[2]

14. M/J/15/21/Q2

Fig. 2.1 shows a student doing a press-up. A total force F acts upwards on his hands. There is also a force upwards on his toes.

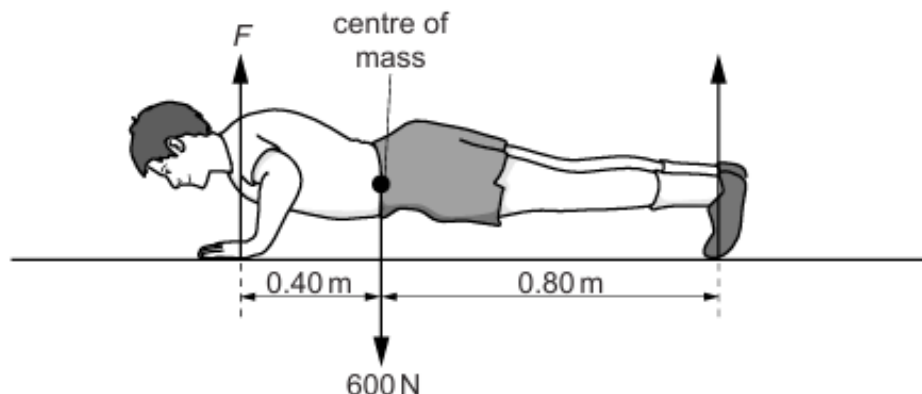


Fig. 2.1 (not to scale)

The weight of the student is 600 N and this force acts downwards from his centre of mass.

- (a) (i) Explain why the student does work as his body rises from the ground.

.....
[1]

- (ii) State the form of energy that the student uses to do this work.

.....[1]

- (b) At the position shown in Fig. 2.1, the student is stationary.
 The weight of the student causes a moment about his toes.

- (i) Calculate the moment of the weight of the student about his toes.

moment =[1]

- (ii) Calculate the value of the force F .

$F =$ [2]

15. O/N/14/22/Q2

In hospitals, doctors and nurses operate taps with their elbows in order to avoid contamination.

Fig. 2.1 shows a hospital tap with a long handle.

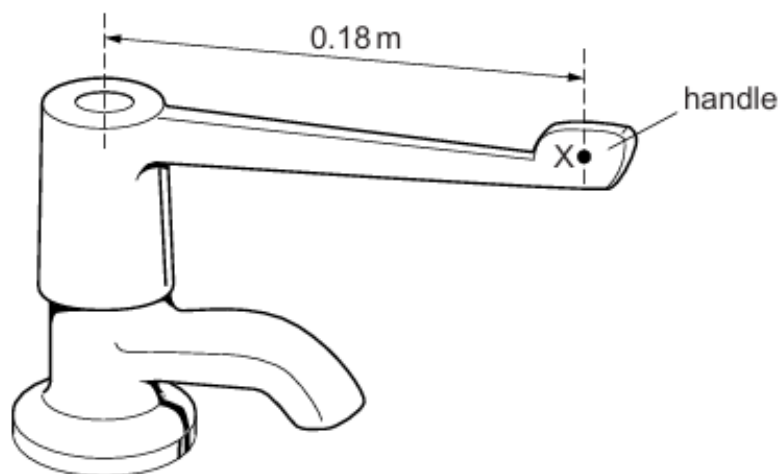


Fig. 2.1

(a) A nurse applies a force of 2.5 N at a point X on the handle, 0.18 m from the axis of the tap.

(i) Calculate the maximum moment about the axis that this force can produce.

moment =[2]

(ii) The moment produced by the nurse is less than this maximum value.

Suggest one reason why this is so.

.....
.....[1]

(b) State how the force needed to operate the tap is affected by the length of the handle.

.....
.....[1]

16. O/N/13/22/Q9

- (a) State what is meant by the *moment of a force*.

.....
[2]

- (b) The anchor of a sailing ship has a mass of 350 kg. Six sailors raise the anchor from the sea-bed by turning a large axle. They push the handles attached to the axle and it rotates. The anchor is on the end of a chain that winds on to the rotating axle.

Fig. 9.1 shows the sailors lifting the anchor.

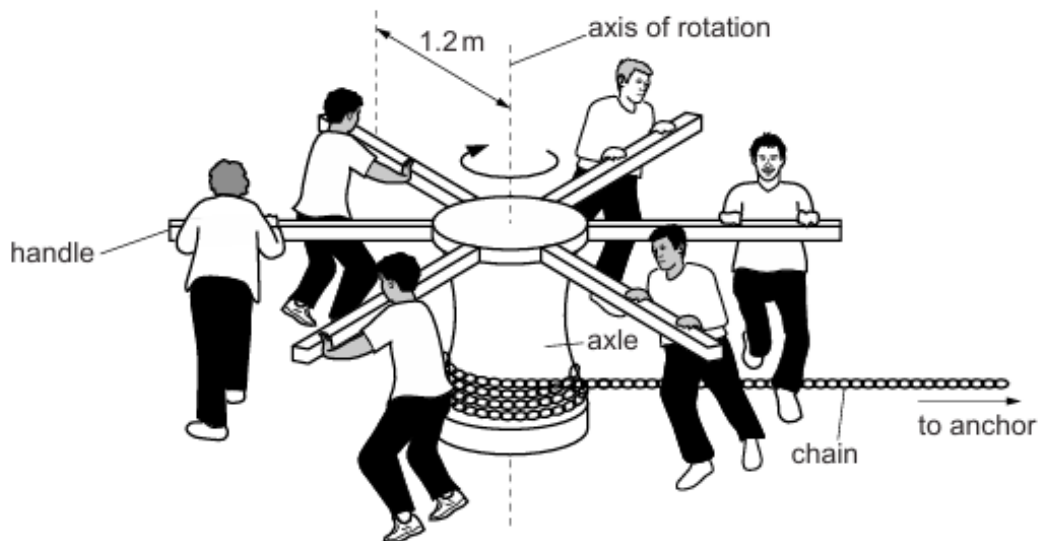


Fig. 9.1

Each of the sailors exerts a force of 750 N on his handle at a distance of 1.2 m from the axis of rotation. The axle rotates through one complete revolution and the anchor is lifted through a distance of 160 cm.

The gravitational field strength g is 10 N/kg.

- (i) Calculate

1. the total moment exerted on the axle by the six sailors,

moment =[2]

2. the gravitational potential energy gained by the anchor as the axle rotates through one complete revolution.

energy =[3]

- (ii) The work done on the axle by the sailors is very much larger than the gravitational potential energy gained by the anchor.

State and explain how energy is wasted.

.....
.....
.....[2]

- (iii) Explain why the power produced by the sailors is larger when the anchor is lifted at a faster rate.

.....
.....[2]

- (c) Describe, with the aid of a labelled diagram, how to verify the principle of moments.

.....
.....
.....
.....
.....
.....[4]

17. O/N/13/21/Q2

A girl of weight 550 N is playing on a see-saw with her brother. Fig. 2.1 shows her brother of weight W sitting 1.1 m to the right of the balance point.

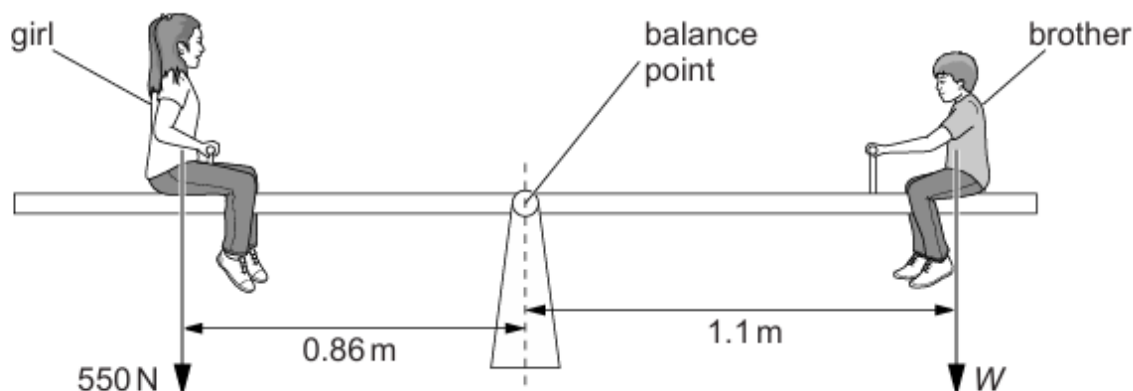


Fig. 2.1 (not to scale)

The see-saw is balanced when the girl sits 0.86 m to the left of the balance point.

(a) Calculate W .

$W = \dots\dots\dots$ [2]

(b) The girl and her brother slide equal distances along the see-saw away from each other.

Describe and explain what happens.

.....

 [3]

18. M/J/13/22/Q2

- (a) Explain what is meant by the *moment of a force*.

.....

 [2]

- (b) Fig. 2.1 shows a system for raising a heavy piece of metal into a vertical position.

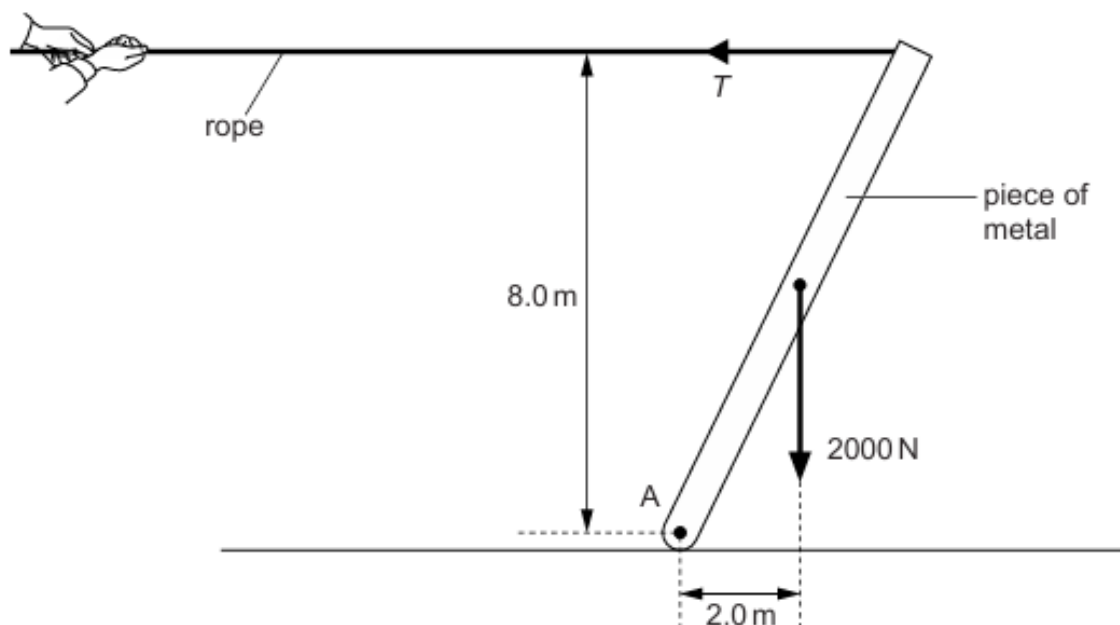


Fig. 2.1 (not to scale)

A man pulls on the rope with a horizontal force T . The piece of metal has a weight of 2000 N and is freely pivoted at A. The system is in equilibrium.

- (i) By taking moments about A, calculate T .

$T =$ [2]

- (ii) The force T and the force that the rope exerts on the man are related by Newton's third law. State the relationship between these forces.

.....

 [2]

19. M/J/13/21/Q2

Fig. 2.1 shows apparatus used to investigate the turning effect of a force.

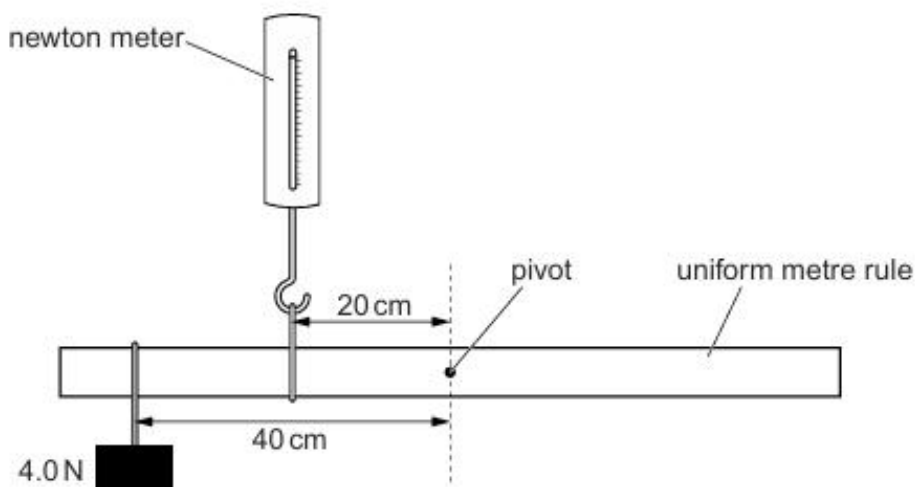


Fig. 2.1

The uniform metre rule is freely pivoted at its centre.

The newton meter is 20 cm from the pivot and a 4.0 N weight is 40 cm from the pivot.

The metre rule is in equilibrium.

(a) State the principle of moments for a body in equilibrium.

.....
 [1]

(b) Calculate the reading on the newton meter.

reading = [2]

(c) The weight of the metre rule is 2.2 N.

Calculate the size of the force exerted on the metre rule by the pivot.

force = [2]

20. O/N/12/22/Q1

- (a) Describe an experiment to verify *the principle of moments*. You may include a diagram in your answer.

.....

.....

.....

.....

.....[4]

- (b) Fig. 1.1 shows a spanner tightening a nut.

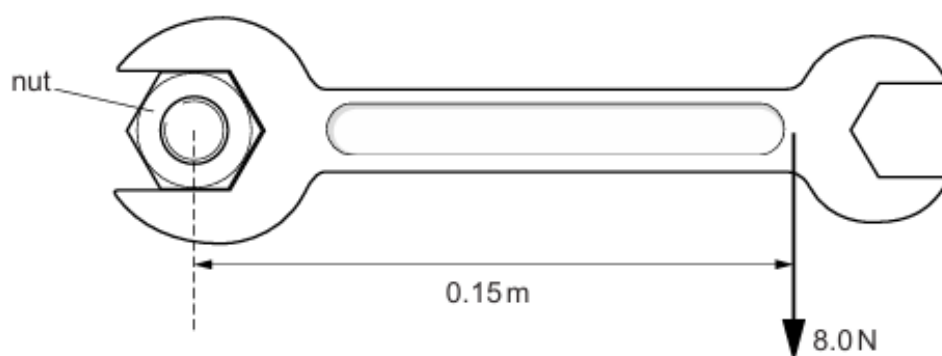


Fig. 1.1

A force of 8.0 N is applied to the spanner at a perpendicular distance of 0.15 m from the centre of the nut.

Calculate the moment of the force acting on the nut.

moment =[2]

21. O/N/11/21/Q11

- A crankshaft is a shaped metal bar that is part of a car engine. It is free to rotate about an axis, as shown in Fig. 11.1.

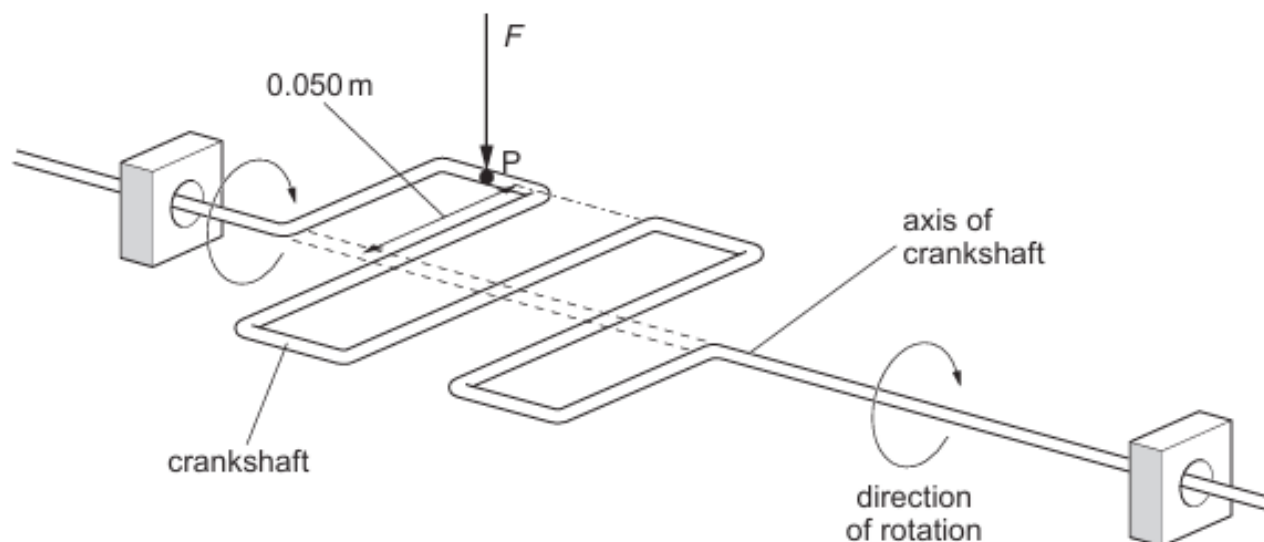


Fig. 11.1

When the crankshaft is horizontal, a vertical force F of 8200 N acts downwards on the crankshaft at P . This causes the crankshaft to rotate. The distance between P and the axis of the crankshaft is 0.050 m .

- (a) (i) State what is meant by the *moment* of a force.

.....
[2]

- (ii) The crankshaft is horizontal. Calculate the moment of F about the axis of the crankshaft.

moment =[2]

- (iii) The size and direction of the vertical force F , acting on the crankshaft at P , remain constant. Explain why the moment of F decreases as the crankshaft rotates through a small angle.

.....
[1]

22. O/N/10/22/Q3

Fig. 3.1 shows a firefighter of total weight 840 N in equilibrium at the top of a ladder that is pivoted at point P.

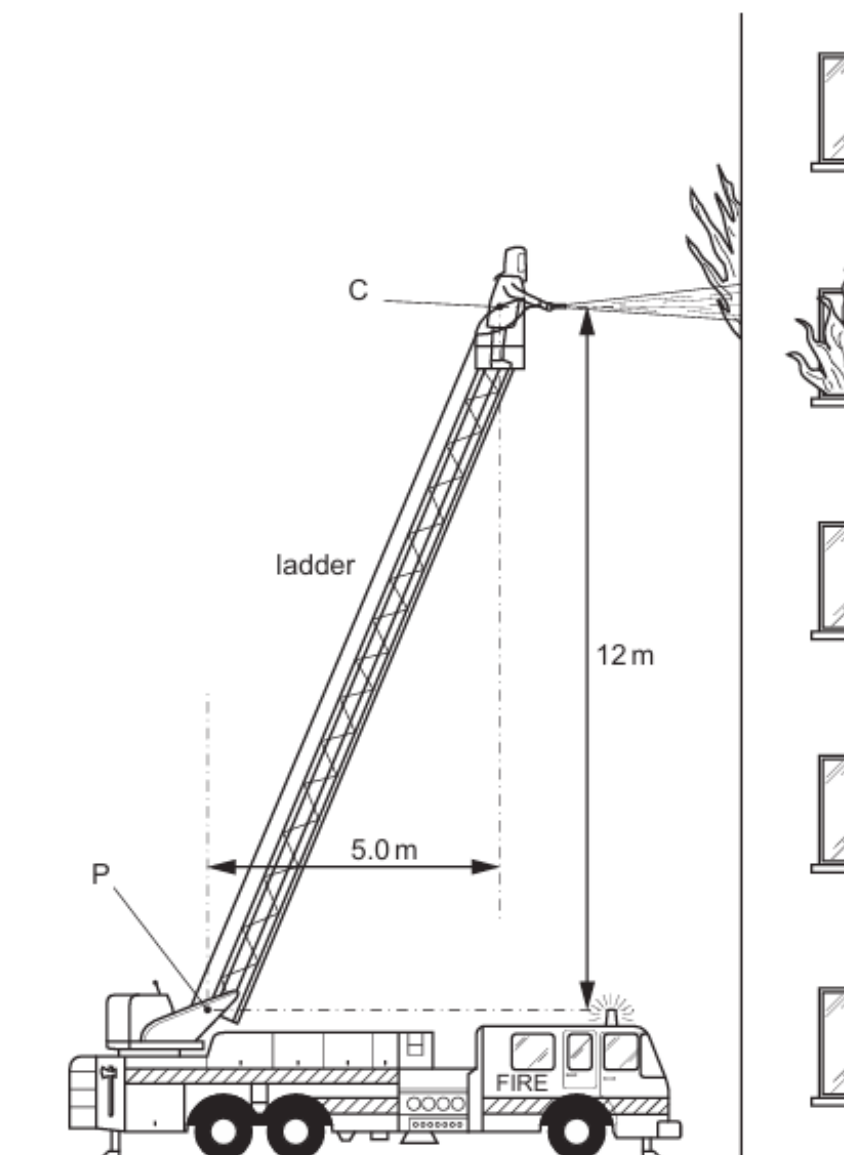


Fig. 3.1

The ladder leans towards a burning building at an angle such that the centre of gravity C of the firefighter is 12 m above and 5.0 m to the right of P. The firefighter holds a hose that directs a high-speed jet of water horizontally into a burning building.

- (a) (i)** Calculate the moment M of the firefighter's weight about P.

moment =[2]

- (ii) The jet of water causes a horizontal force R on the firefighter that acts towards the left, through C. This opposes the turning effect of his weight. Calculate the size of R that, on its own, ensures that M is exactly cancelled.

force =[1]

- (iii) Suggest a third force that has a turning effect about P on the ladder.

.....
.....[1]

23. O/N/09/Q7

Fig. 7.1 shows a rectangular concrete slab of weight 18000 N. It rests on a brick wall and is the roof of a bus shelter. The concrete slab is 3.0 m wide.

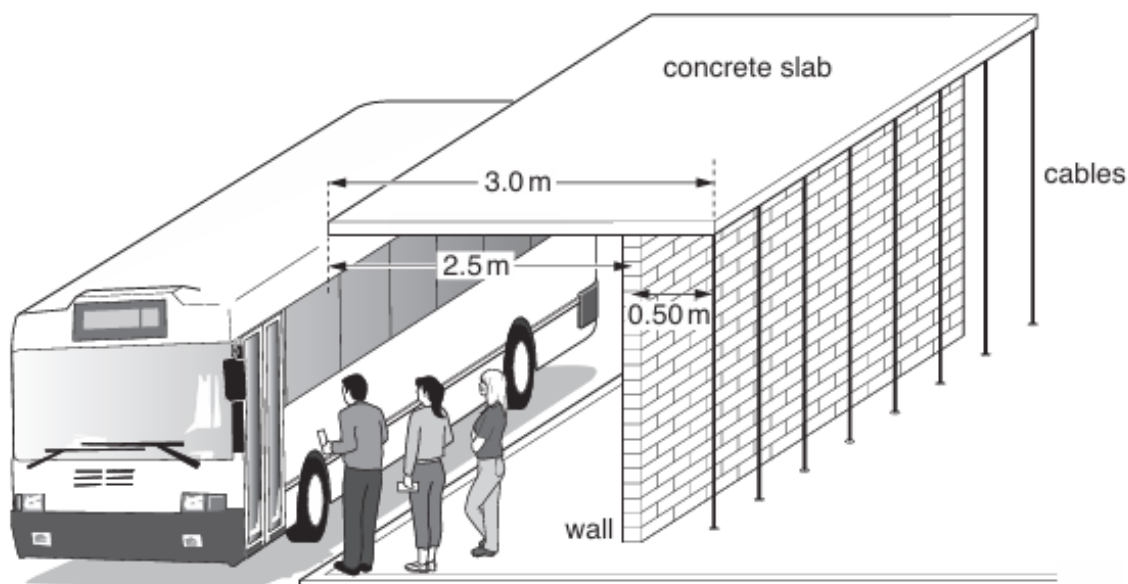


Fig. 7.1

The wall is 2.5 m from the front of the concrete slab and 0.50 m from the back. The cables behind the shelter pull downwards and stop the slab toppling forwards.

- (a) The concrete slab is of uniform thickness and density. Determine the perpendicular distance between the wall and the centre of mass of the slab.

distance = [1]

- (b) (i) State the principle of moments.

.....

 [2]

- (ii) Calculate the total downward force exerted by the cables on the slab.

force = [2]